

◆ **Zoho-Focused Important Algorithms List**

1. Sorting & Searching

- Bubble Sort, Selection Sort, Insertion Sort (basic expected)
 - Merge Sort & Quick Sort (divide and conquer must-know)
 - Binary Search (standard + variations: first/last occurrence, floor/ceil)
 - Two-Pointer Technique (pairs, triplets, sliding windows)
-

2. String Algorithms

- Palindrome check, Longest Palindromic Substring
- Substring Search (Naive, KMP expected at advanced level)
- Anagram check & Grouping
- Longest Common Prefix (for multiple strings)
- Remove Consecutive Duplicates
- Minimum Window Substring (sliding window)
- Check Rotation of Strings
- Pattern Matching (without built-ins)

3. Recursion & Backtracking

- Generating Permutations & Combinations
- N-Queens Problem
- Rat in a Maze / Grid Path finding
- Subset Sum / Power Set Generation

4. Dynamic Programming (DP)

- Fibonacci (recursive → DP optimized)
- Longest Common Subsequence (LCS)
- Longest Increasing Subsequence (LIS)
- Minimum Edit Distance
- Coin Change Problem

5. Array & Matrix Based

- Kadane's Algorithm (Maximum Subarray Sum)
- Rotate Matrix (90° in place)

- Spiral Traversal of Matrix
 - Trapping Rainwater Problem
 - Subarray with Given Sum (sliding window + prefix sum)
 - Leaders in Array (right to left scan)
-

6. Graph & Tree Basics

- Depth-First Search (DFS)
 - Breadth-First Search (BFS)
 - Detect Cycle in Graph (DFS/Union-Find)
 - Minimum Spanning Tree (Prim's/Kruskal's basics) – sometimes asked in advanced rounds
 - Binary Tree Traversals (Inorder, Preorder, Postorder, Level Order)
-

7. Greedy Algorithms

- Activity Selection Problem
- Huffman Coding (rare, but asked in high-level rounds)
- Job Scheduling with Deadlines

8. Other Important Ones

- Hashing-based problems (frequency maps, duplicates check)
- Stack-based algorithms (Next Greater Element, Balanced Parentheses)
- Queue-based problems (Sliding Window Maximum)
- Bit Manipulation (XOR-based unique element, subset generation)